The main entities we'll focus on are:

- **University**
- **Faculty**
- **Department**
- **Course**
- **Student**

Step 1: Identify Entities and Their Attributes

First, determine the attributes for each entity.

1. **University**
    - university_id (Primary Key)
    - university_name
    - Address
2. **Faculty**
    - faculty_id (Primary Key)
    - faculty_name
    - HiringDate
    - university_id (Foreign Key)
3. **Department**
    - department_id (Primary Key)
    - department_name
    - Campus
    - Address
    - faculty_id (Foreign Key)
4. **Course**
    - Course_id (Primary Key)
    - Course_name
    - Credits
    - Format
    - Course Level

- Prerequisites
- SeatsAvalible
- department_id (Foreign Key)

5. **Student**
   - student_id (Primary Key)
   - first_name
   - last_name
   - email
   - Age
   - HomeAddress
   - SportsType
   - department_id (Foreign Key)
6. **Enrollment** (Associative Entity)
   - enrollment_id (Primary Key)
   - student_id (Foreign Key)
   - course_id (Foreign Key)
   - enrollment_date

Step 2: Determine Relationships Between Entities

- A **University** has many **Faculties**.
- A **Faculty** belongs to one **University** and has many **Departments**.
- A **Department** belongs to one **Faculty** and offers many **Courses**.
- A **Student** belongs to one **Department** and can enroll in many **Courses**.
- A **Course** is offered by one **Department** and can have many **Students** enrolled.

Step 3: Create the Database Schema

Using SQL `CREATE TABLE` statements, we'll define the tables and their relationships.

Step 4: Insert Sample Data

Step 5: Retrieve Information from the Tables

Now, let's perform some queries to retrieve information.

1. **List all students enrolled in 'Introduction to Programming'**
2. **Get all courses offered by the 'Computer Science' department**
3. **Find the total number of students in each department**
4. **Retrieve all faculties within 'Global University**
5. **List all courses a specific student is enrolled in**

Step 6: Update and Delete Operations (Optional)

You may also need to perform update and delete operations.

- **Update a student's email**
- **Delete a course from the system**

Step 7: Advanced Queries

1. **Find students who are enrolled in courses outside their department**
2. **Get the list of departments and the number of courses they offer**

Step 8: Practice Exercises

- **Exercise 1:** Write a query to find all students who are not enrolled in any courses.

Hint: Use a LEFT JOIN between Student and Enrollment and look for NULL in Enrollment.

- **Exercise 2:** List all faculties and the departments under them.

Hint: Use a JOIN between Faculty and Department.

- **Exercise 3:** Find the average number of credits for courses in each department.

***Hint**: Use AVG aggregate function and GROUP BY department_id.*

Conclusion

This step-by-step guide provides a foundational approach to designing a course registration system using SQL. By following these steps, you'll understand how to model entities, define relationships, and manipulate data using SQL commands. Practice by extending the system with additional entities like professors, classrooms, or schedules to enhance your learning experience.